UNIT-2 Nuclear Energy

Fundamental forces in the universe :- Fundamental forces govern the behavior of matter and energy in the universe and are essential for understanding the structure and interactions of all particles and objects.

These forces in the universe are:

1. **Gravitational force** :- This force is responsible for the attraction between objects with mass. It acts over large distances and is the weakest of the four fundamental forces.

2. **Electromagnetic force** :- This force is responsible for electric and magnetic effects between charged particles. It is much stronger than gravity and acts over both long and short distances.

3. **Strong Nuclear force** :- This force binds protons and neutrons together in the nucleus of an atom. It is very strong but acts over extremely short distances, binding quarks together to form protons and neutrons.

4. **Weak Nuclear force** :- This force is responsible for radioactive decay and certain type of nuclear reactions. It is much weaker than the strong nuclear force and has a very short range.

Quantum mechanics relevent for nuclear Physics

Quantum mechanics plays a crucial role in nuclear physics, especially in understanding the behavior and interactions of particles within atomic nuclei.

Here are some key points:

1. Nuclear structure and Binding :- Quantum mechanics explains how protons and neutrons (collectively called nucleons) are bound together within atomic nuclei. It describes how these particles interect via stong nuclear force.

2. Shell structure :- Similar to electrons in atoms, nucleous occupy discrete energy levels or shells within the nucleous. The shell model, derived from quantum mechanics, helps in understanding nuclear stability, magic numbers and isotopic properties.

3. Nuclear Reactions :- Quantum mechanics governs the probabilities and outcomes of nuclear reactions, including radioactive decay, fusion and fission. It also predicts reaction rates, energy distributions of emitted particles.

4. Experimental Techniques :- Quantum mechanics guides the interpretation of experimental data from nuclear physics experiments, providing theoritical and numerical framework. This helps for designing nuclear reactors.

Nuclear Forces:- Nuclear forces are the forces that hold together the particles (Protons & Neutrons) within an atomic nucleus.

These forces are responsible for the stability of atomic nucleus and the behavior of nuclear matter.

There are two primary types of nuclear forces:

1. **Strong Nuclear Force:-** This is the force that binds protons and neutrons together in the nucleus.

2. **Weak Nuclear Force:-** This force is responsible for certain types of radioactive decay such as beta decay, where a neutron decays into a proton, electron and antineutrino.

Properties of Nuclear Force

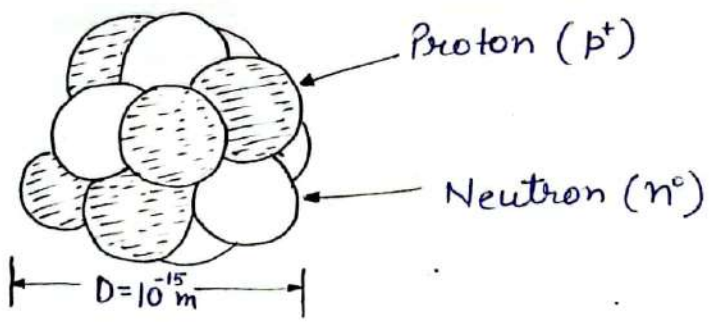
1. The nuclear force acts over a very short range, typically about 1 to 2 femtometers (10^{-15} to 10^{-14} meters).
2. This force is primarily attractive between nucleons.
3. Nuclear forces are responsible for the binding energy that holds the nucleus together.
4. This force is approximately charge independent meaning it affects protons and neutrons equally.
5. This force determine the structure of atomic nucleus, including their size, shape and stability.

Energy scale of Nuclear Forces

The Nuclear force operates on an energy scale of around 1 to 10 MeV (Million electron volts).
This energy scale is derived from the masses of the particles involved in nuclear interactions and determines the binding energy of ~~nu~~ nucleons within atomic nuclei.

Nuclear Structure :-

- Nuclear structure refers to the arrangement and organization of protons and neutrons within an atomic nucleus.
- The nucleus is less than one ten-thousandth the size of the atom.
- The nucleus contains more than 99.9% of the mass of the atom.

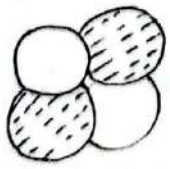


- The diameter of nucleus approximately is 10^{-15} m.

Radioactivity, Nuclear decays

- Radioactivity refers to the spontaneous emission of radiation from the nucleus of an atom.
- This phenomena occurs when the nucleus is unstable and undergoes radioactive decay to achieve a more stable configuration.
- Radioactivity is also known as radioactive decay, nuclear decay, radioactive disintegration or nuclear disintegration.
- For example, uranium and thorium are two radioactive elements found naturally in the Earth's crust. Over billions of years, these two elements slowly change form and produce decay products such as radium and radon.

Alpha Decay :- In alpha decay, a positively charged particle, identical to the nucleus of helium 4, is emitted spontaneously. This particle, also known as an alpha particle, consisting two protons and two neutrons.
It was discovered by Sir Ernest Rutherford in 1899.



α (alpha particle) = $_2He^4$

- Plutonium 239 decays by alpha particle emission as follows:

$$_{94}Pu^{239} \xrightarrow[24,000 \text{ years}]{} {}_{92}U^{235} + {}_{2}He^{4}$$

- Alpha decay usually occurs in heavy nuclei such as uranium or plutonium, and therefore is a major part of the radioactive fall out from a nuclear explosion.
- Since an alpha particle is relatively more massive than other forms of radio active decay, it can be stopped by a sheet of paper and can not penetrate human skin.
- A 4 MeV alpha particle can only travel about 1 inch through the air.

<u>Beta decay</u>:- Atoms emit beta particles through a process known as beta decay.

Beta decay occurs when an atom has either two many protons or too many neutrons in its nucleus.

Two types of beta decay can occur.

- One type (positive beta decay) releases a positively charged beta particle called a positron, and a neutrino.

- The other type (negative beta decay) releases a negatively charged beta particle called an electron, and an antineutrino.
- The neutrino and the antineutrino are high energy elementary particles with little or no mass.
- These particles are released in order to conserve energy during the decay process.
- Negative beta decay is far more common than positive beta decay.

β (beta particle) $= {}_{-1}e^{0}$

The tritium beta-decay process is written as follows:

$$ {}_{1}H^{3} \xrightarrow[12.3 \text{ years}]{} {}_{2}He^{3} + {}_{-1}e^{0} + {}_{0}\bar{\nu}^{0} $$

(beta particle) (antineutrino)

- This form of radioactive decay was discovered by Sir Ernest Rutherford in 1899, although the neutrino was not observed until the 1960s.

- Beta particles ~~ar~~ have all the characteristics of electrons.
- At the time of their emission, they travel at nearly the speed of light.
- A typical 0.5 MeV particle will travel about 10 feet through the air, and can be stopped by 1-2 inches of wood.

<u>Gamma decay</u> :- Gamma decay is a type of radio active decay that occurs when an excited nucleus emits a gamma ray photon.

- The time it takes for this process to occur is usually very short, typically around 10^{-12} seconds.
- Unlike alpha or beta decay, gamma decay does not change the number of protons or neutrons in the nucleus; it only releases energy, bringing the nucleus from a higher energy state to a lower energy state.

γ (Gamma rays) = γ-radiation

(i) $$_{53}I^{125*} \longrightarrow {}_{53}I^{125} + {}_{0}\gamma^{0}$$

Higher Energy Level

Lower Energy Level

Energy (γ-rays)

(ii) $$_{90}Th^{234*} \longrightarrow {}_{90}Th^{234} + {}_{0}\gamma^{0}$$

(iii) $$_{2}He^{3} \longrightarrow {}_{2}He^{3} + {}_{0}\gamma^{0}$$

- Gamma decay typically occurs after other types of decay, where the nucleus is left in an excited state. The nucleus then releases this excess energy as gamma radiation to reach a more stable (lower energy) state.
- Gamma rays have a high penetrating ability and can pass through most materials, including human tissue. Dense materials like lead or thick layers of concrete are required to shield against gamma radiation effectively.

Applications of α, β & γ-decay

(i) Alpha decay: Used in smoke detectors, cancer treatment and radiometric dating.

(ii) Beta decay: Used in medical imaging and treatment, radiometric dating & industrial quality control.

(iii) Gamma decay: Used in medical imaging and treatment, industrial radiography, sterilization and scientific research.

Half Life :- The half life of a radioactive substance is the time required for half of the atoms in a sample to undergo radioactive decay. This concept is fundamental in understanding how radioactive materials change over time.

For example, if a substance has a half life of 10 years, after 10 years only half of the initial amount will remain; after 20 years, only a quarter of the initial amount will remain, and so forth.

The half-life concept is used to:

1- Determine decay rates of radioactive substances.

2- Date ancient material using carbon-dating.

3- Plan radiotherapy treatment in cancer.

4- Assess environmental impact of pollutants.

5- Plan drug dosing based on how drugs remain active in the body.

<u>Nuclear Reactions</u> :- Nuclear reactions involve changes in the nucleus of an atom and can be classified into several types:

1- Fission

2- Fusion

3- α- decay

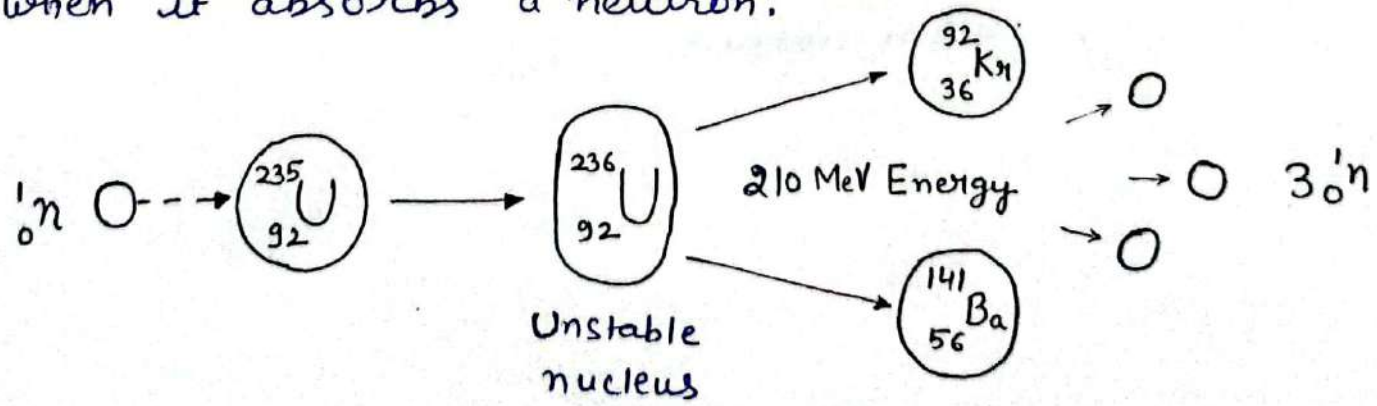
4- β- decay

5- γ- decay

<u>Nuclear Fission</u> :- A heavy nucleus splits into smaller nucleus, releasing a large amount of energy.

This process is used in nuclear reactors and atomic bombs.

For example, Uranium -235 can undergo fission when it absorbs a neutron.



- The equation of this reaction has been given below:

$$\left\{ {}^{235}_{92}U + {}^{1}_{0}n \longrightarrow {}^{141}_{56}Ba + {}^{92}_{36}Kr + 3\,{}^{1}_{0}n + 210\,MeV\ Energy \right\}$$

- The other example of nuclear fission is splitting of <u>Uranium -233</u>

$$\left\{ {}^{233}_{92}U + {}^{1}_{0}n \longrightarrow {}^{137}_{54}Xe + {}^{94}_{38}Sr + 3\,{}^{1}_{0}n + Energy \right\}$$

- Splitting of <u>Plutoniam -239</u>

$$\left\{ {}^{239}_{94}Pu + {}^{1}_{0}n \longrightarrow {}^{137}_{54}Xe + {}^{40}_{103}Zr + 3\,{}^{1}_{0}n + Energy. \right\}$$

<u>Note:-</u> Ba → Barium Sr → Strontium

Kr → Krypton Zr → Zirconium

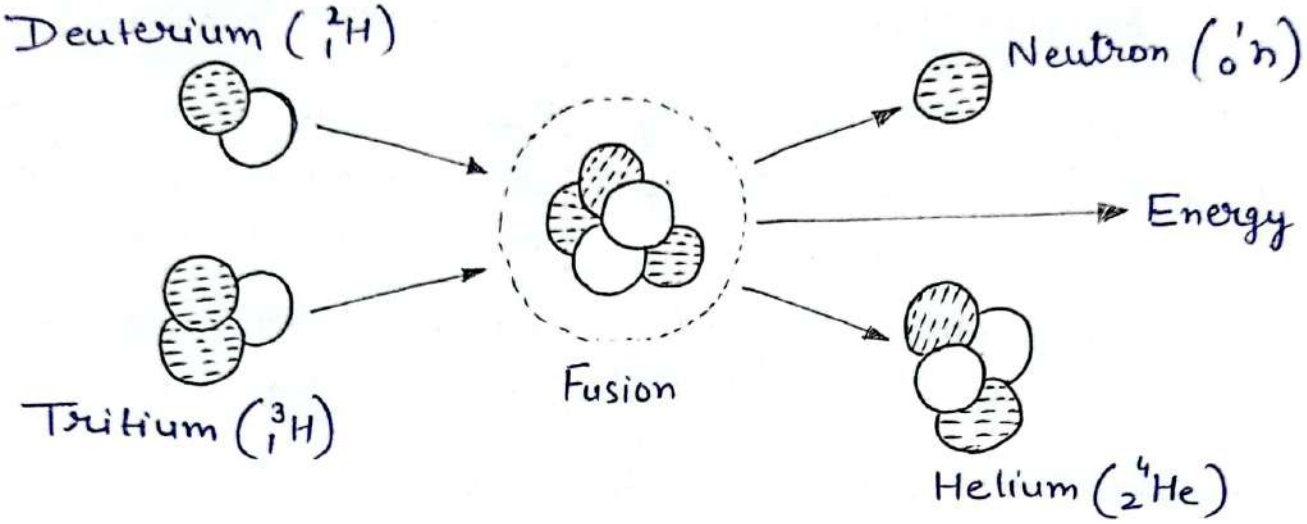
Xe → Xenon

<u>Nuclear Fusion</u> :- Two light nucleus combine to form a heavier nucleus, releasing energy. —

This process powers stars, including the sun, and is the basis for hydrogen bombs.

For example, hydrogen isotopes deuterium and tritium fuse to form helium.

- This process releases energy because the resulting nucleus has less mass than two original nucleus, and the leftover mass is converted into energy.



- The equation of further reaction is:

$$^{2}_{1}H + ^{3}_{1}H \longrightarrow ^{4}_{2}He + ^{1}_{0}n + 17.59 \text{ MeV Energy.}$$

Differences between Nuclear fission and fusion

	Nuclear Fission	Nuclear Fusion
Definition	• Fission is the spliting of a large atom into two or more smaller ones.	• Fusion is the combining of two or more lighter atoms into a larger one.
Occurrence	• Fission reaction does not normally occur in nature.	• Fusion occurs in stars, such as the Sun.
Energy Requirement	• Takes little energy to split two atoms in a fission reaction.	• Extremely high energy is required to fuse atoms in fusion reaction
Energy Released	• The energy released by fission is million times greater than that released in chemical reaction.	• The energy released by fusion is three to four times greater than fission.
Energy Production	• Fission is used in nuclear Power Plant.	• Fusion is an experimental technology for producing power.

Chain Reaction :- A process in reaction components undergo multiple reactions until an end point occurs is called a chain reaction.

Fire is an example of a chain reaction that will continue until all reaction material is usedup or an end point of putting out the fire occurs.

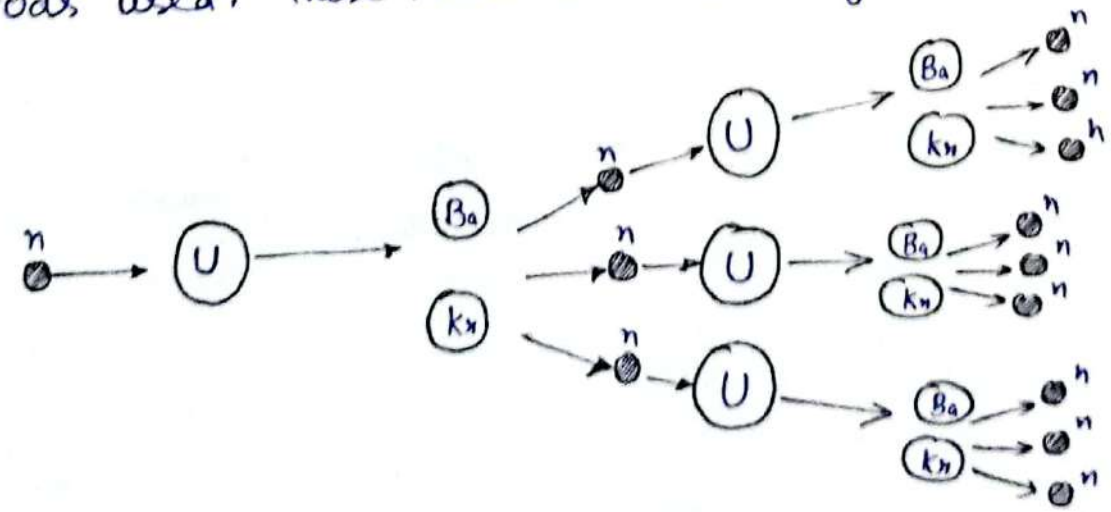
Radical reactions and nuclear fission are two types of chain reactions that involve repeated reactions.

Nuclear Chain Reactions :- A nuclear chain reaction refers to a process in which neutrons released in fission, produce an additional fission in atleast one further nucleus. This nucleus in turn produces neutrons, and the process repeats.

The process may be controlled (nuclear power plant) or uncontrolled (nuclear weapons).

All neutrons in an uncontrolled chain reaction are allowed to induced fission reactions.

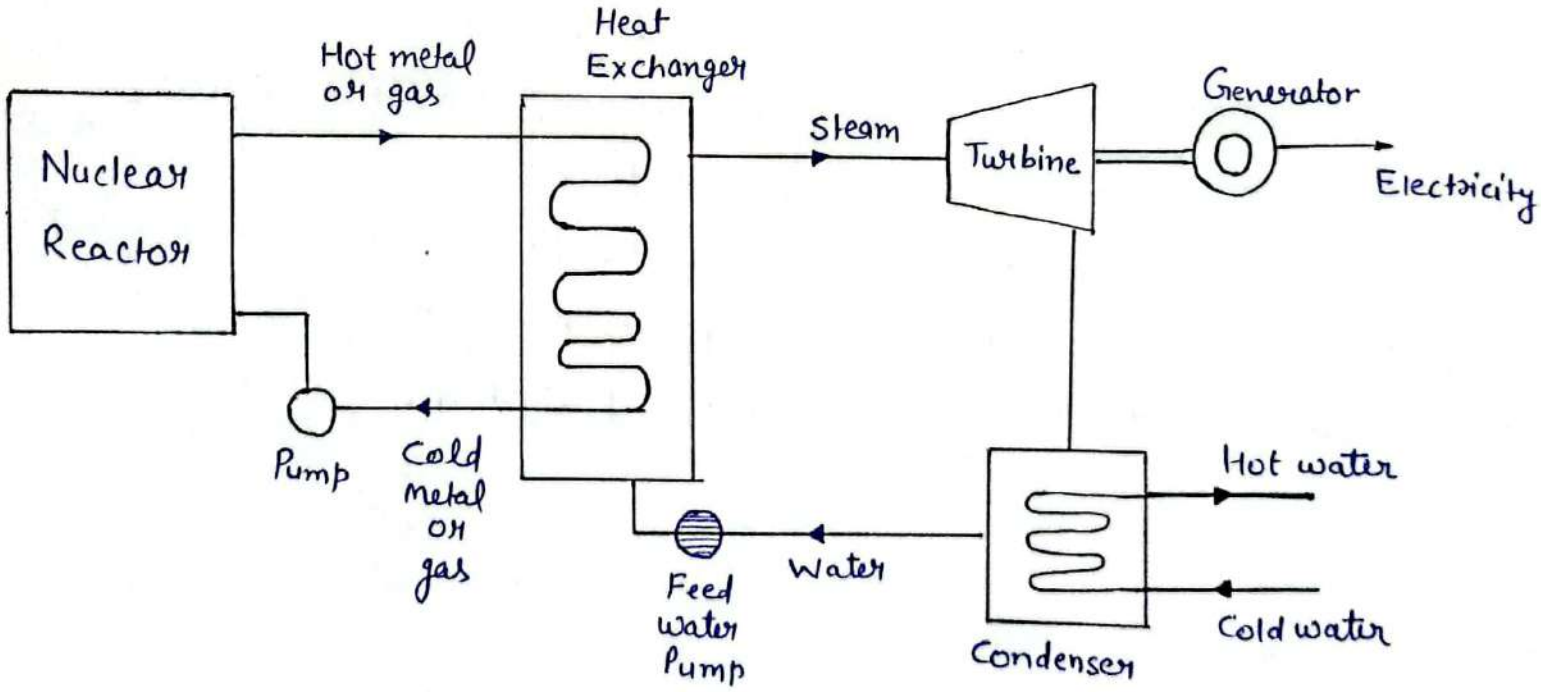
The number of neutrons allowed to do this in a Controlled chain reaction is determined by the number of control rods used. These rods are made of Cadmium or boron.



Nuclear Power Plant :- A nuclear power plant generates electricity through nuclear fission, a process where atomic nucleus split and release energy in nuclear reactor. This energy heats water to produce steam, which drives turbines connected to generators.

Nuclear power plants are known for their ability to produce large amount of electricity with low green-house gas emissions.

However, they also come with challenges, including managing radioactive waste, ensuring safety and dealing with the high costs of construction and decommissioning.



Nuclear Reactor

A nuclear reactor is a device used to initiate and control a sustained nuclear chain reaction, typically for generating electricity. It works by splitting heavy atomic nucleus such as uranium-235 or plutonium-239, in a process called nuclear fission.

Here's a basic outline of its key components and functions:

1. Fuel Rods :- These rods contain the nuclear fuel (usually uranium or plutonium) that undergoes fission.

2. Control Rods :- These rods made of materials (Boron, Cadmium, Hafnium alloy) that absorb neutrons. These rods regulate the fission process by controlling the number of neutrons available to sustain the chain reaction.

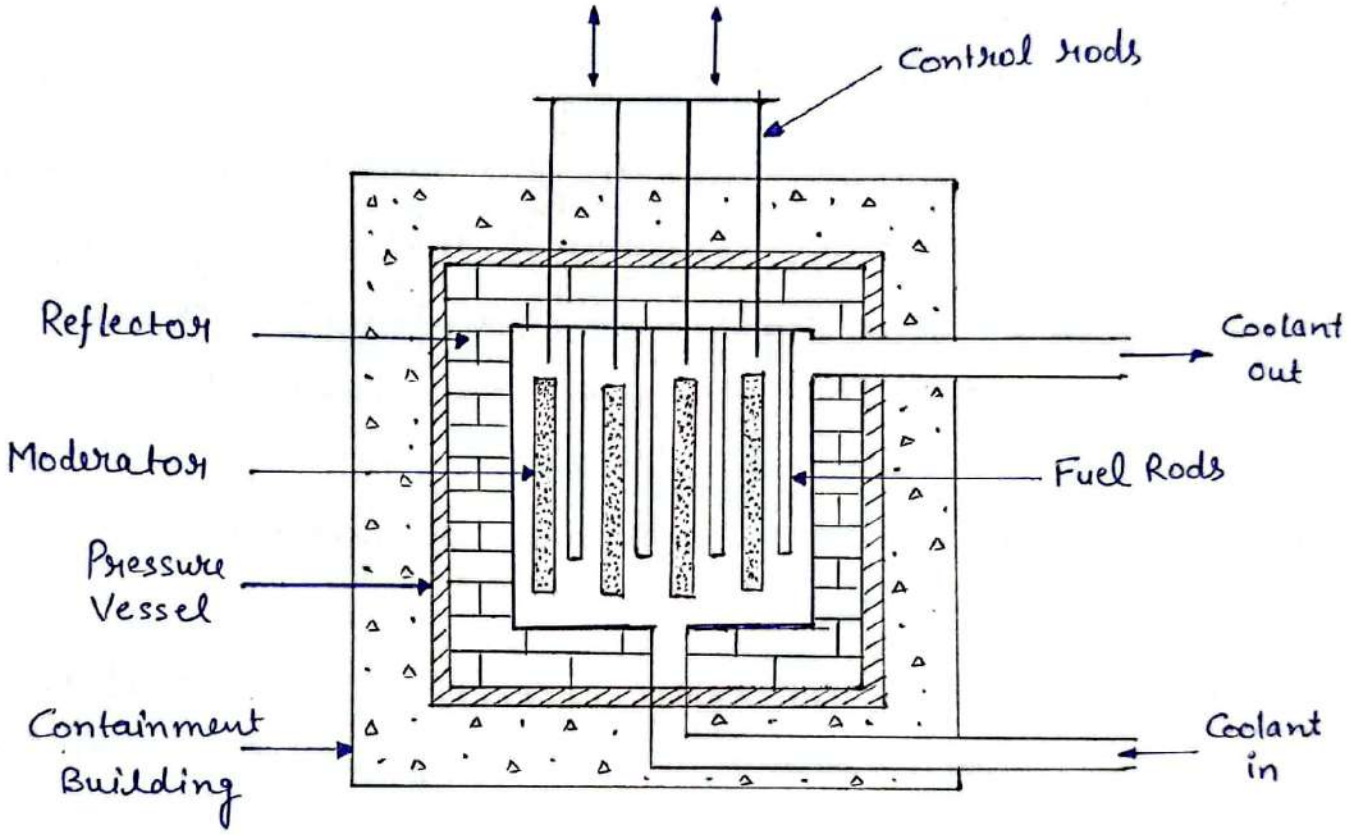
3. Moderators :- A substance, often water or graphite, slows down the speed of neutrons produced during fission to sustain the chain reaction efficiently.

4. Coolant :- A fluid (often water) circulates through the reactor to absorb the heat produced by fission. The heated coolant then transfer energy to a heat exchanger or steam generator.

5. Pressure Vessel :→ The reactor core, where the fission reactions occur, is contained within a robust pressure vessel to handle the high pressure and temperature.

6. Containment Building :- A reinforced structure surrounding the reactor vessel designed to contain any potential release of radioactive material.

7. <u>Reflector</u> :- It is a structure that surrounds the reactor core and serves to reflect neutrons back into the core.



Types of Nuclear Reactor

1. Pressurized Water Reactor (PWR)

- Cooling System: Uses water under high pressure to cool the reactor core. The high pressure prevents the water from boiling even at high temperature.
- Operation: The pressurized water transfers heat to a secondary loop through a heat exchanger. where the water in the secondary loop is allowed to boil and produce steam to drive turbine.
- Advantages: Generally has a stable operation and high efficiency. It is widely used, including in many commercial power plants

2. Boiling water Reactor (BWR)

- Cooling System: Uses water that boils directly in the reactor core to produce steam.
- Operation: The steam generated in the reactor core is used directly to drive turbines. The steam is then condensed back into water and recirculated.
- Advantages: Simplifies the reactor design by eliminating the need of a separate heat exchanger.

3. Fast Breeder Reactor (FBR)

- Cooling System: Uses liquid metal, such as sodium or lead, as the coolant because these metals have excellent thermal conductivity and can operate at high temperatures.
- Operation: FBRs use fast neutrons (as opposed to thermal neutrons used in PWRs and BWRs) to sustain the nuclear chain reaction. They are designed to breed more fissile material than they consume.
- Advantages: Can utilize a wider range of nuclear fuels and generate more fuel than they consume, thus potentially extending the availability of nuclear fuel resources.

4. Canada Deuterium Uranium (CANDU) Reactor
OR
Heavy water Reactor (HWR)

- Cooling System: Heavy water (deuterium oxide, D_2O)
- Operation: These reactors used natural uranium as fuel and heavy water as both a moderator and coolant to cool reactor core.
- Advantages: They have a strong safety record and are known for their flexibility in fuel use.

These are used in Canada, South Korea & Romania.

Note: Some other types of nuclear reactors are:
- Graphite moderated Reactor (GMR)
- Molten Salt Reactor (MSR)
- High temperature-Gas cooled Reactor (HTGR)

Binding Energy :- The energy required to separate a nucleus into its individual protons and neutrons. It is also equal to the energy released when these nucleons (protons and neutrons) come together to form the nucleus.

Calculation : The binding energy can be calculated using the mass energy equivalence principle.

where the difference between the mass of the nucleus and the sum of the masses of its constituent nucleons is converted into energy using Einstein's equation.

$$\{ \Delta E = \Delta m \, c^2 \} \text{ Joule}$$

where

$$\left\{ \begin{array}{l} \Delta m = \text{mass defect} \\ \quad \text{or} \\ \Delta m = m_{nucleus} - m_{nucleons} \end{array} \right\}$$

and

$$\left\{ \begin{array}{l} c = \text{speed of light} \\ \quad \text{or} \\ c = 3 \times 10^8 \text{ m/s} \end{array} \right\}$$

$\because$ 1 Joule $= 6.242 \times 10^{12}$ MeV

$\therefore$ we can calculate the binding energy in MeV

$$\{ \Delta E = \Delta m \times 931.5 \text{ MeV} \}$$

Binding Energy per Nucleon

This is the binding energy divided by the total number of nucleons (protons + neutrons) in the nucleus.

It provides a measure of the average energy binding each nucleon within the nucleus.

Significance: It helps in comparing the stability of different nucleus. Generally nucleus with higher energy per nucleon are more stable

Que:- Calculate the binding energy and binding energy per nucleon (in MeV) of a Nitrogen nucleus $_7N^{14}$ from the following data:

mass of proton = 1.00783 u

mass of neutron = 1.00867 u

mass of Nitrogen nucleus = 14.00307 u

Sol^n:- (i) Total number of nucleons = Number of Protons + Number of neutrons

$= 7 + 7 = 14$

(ii) mass defect $\Delta m = m_{nucleons} - m_{nucleus}$

$= 7\left(\text{mass of proton}\right) + 7\left(\text{mass of neutron}\right) - \text{mass of nucleus}$

$= 7(1.00783) + 7(1.00867) - 14.00307$

$\Delta m = 0.11243$ amu

(iii) Binding energy $\Delta E = \Delta m \times 931.5$ MeV

$= 0.11243 \times 931.5$

$\{\Delta E = 104.73 \text{ MeV}\}$

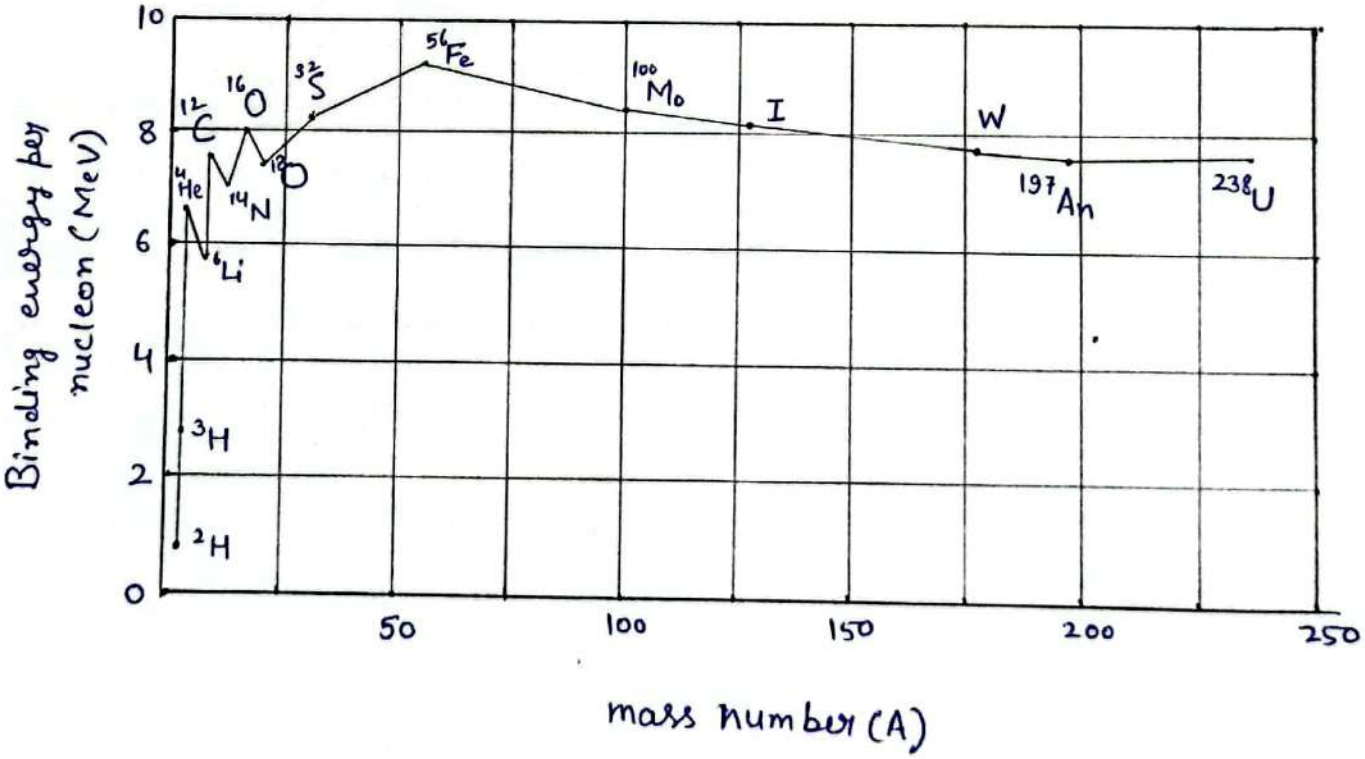
(iv) Binding energy per nucleon = $\frac{\text{Binding energy}}{\text{Total Number of nucleons}}$

$$= \frac{\Delta E}{14}$$

$$= \frac{104.73}{14}$$

{ Binding energy per nucleon = 7.48 MeV }

<u>Binding Energy Curve</u> :- This curve is also known as the binding energy per nucleon curve. This curve shows that binding energy per nucleon the stability of different nucleus of different mass number (A).



Unit -2 Important Questions and PYQ (AKTU)

1- What are four fundamental forces in the Universe? Explain

2- What do you mean by nuclear forces? what are their types? Explain briefly.

3- What is the importance of quantum mechanics? what are some useful applications of nuclear physics?

4- What do you mean by binding energy? Draw the binding energy curve.

5- Differentiate between fusion and fission nuclear reaction.

6- What do you mean by Half life and mean life?

7- What is chain reaction in nuclear fission process?

8- Explain pressurized water reactor (PWR).

9- Explain boiling water reactor (BWR).

10- What is the safest nuclear reactor disign? What are the four main components of a fission reactor?

11- Define fast breeder reactors.

12- Write down the advantages of nuclear energy.

13- What is radioactive decay? Explain their types.

14- Calculate the binding energy per nucleon of

(i) ${}_{6}C^{12}$ (ii) ${}_{17}Cl^{35}$